PAPER DEMONSTRATION

SYNAPSE-S2

Visual Operator Manual

A guided tour of the Namespace Galaxy, cortex, ganglia, neurons, governed recall, memory maintenance, and reliability controls.

THE CORE IDEA

The Galaxy is a read-only neural atlas. Use it to locate and understand memory. Use Cortex Governor, Memory Write, Recall, Memory Graph, and governed bridge controls to change durable state.

READ-ONLY ATLAS | GOVERNED WRITES | EXPLICIT PRUNE | VERIFIED RECOVERY

Interface shown: release 018cd32db4013dee7763f1517f953eb956b7fa0d

Live dashboard examples supplied July 29, 2026

Budget proposal supporting document

01 - READ THE SYSTEM CORRECTLY

The memory anatomy

READ-ONLY ATLAS

A visual level of detail is not a mutation boundary. Each inward step reveals more structure while preserving the same durable namespace.

VISUAL MASS FORMULAS

Galaxy

58% memory volume
27% indexed density
15% approved bridges

Cortex

72% bounded log memories
28% relationships per memory

Ganglion

68% relative log memories
32% weighted edge density

Neuron

Relative log visible weighted degree
inside the returned sample

WHAT SIZE MEANS

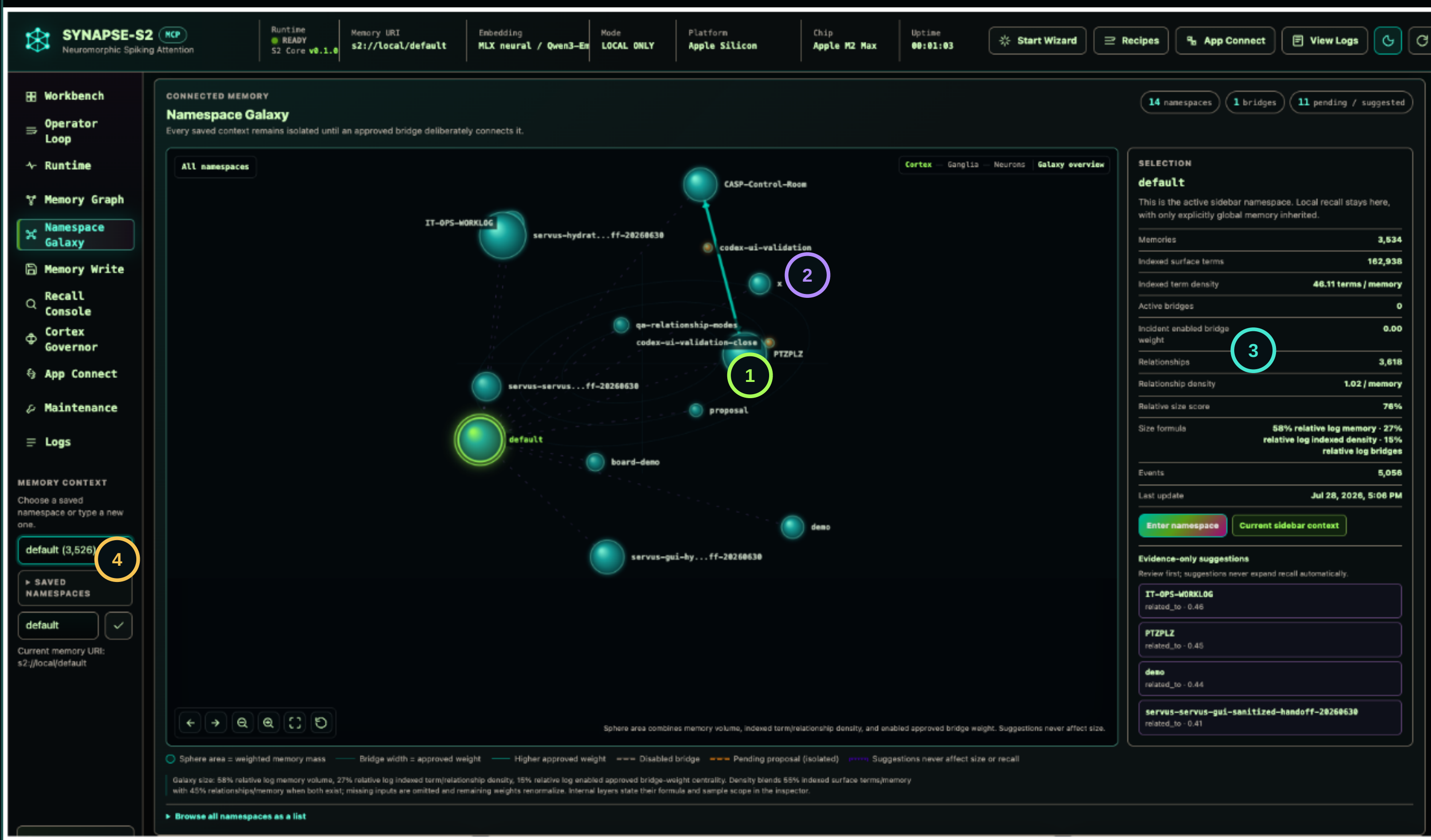
- A relative visual comparison of stored volume and connectivity.
- A fast way to spot dense or highly connected areas.
- A navigational cue that changes as the selected scope changes.

WHAT SIZE NEVER MEANS

- Truth, confidence, importance, quality, recency, or permission.
- A command to retain, delete, promote, or bridge memory.
- Proof that every relationship in the database is currently visible.

All namespaces: the Galaxy

READ-ONLY + SCOPE



1 Namespace spheres

Sphere area combines memory volume, indexed density, and enabled approved bridge centrality.

2 Bridge and suggestion lines

Only approved enabled bridges grant Connected recall. Suggestions stay isolated.

3 Selection inspector

Read totals, density, formula, update time, and bridge evidence before acting.

4 Active sidebar context

This scope governs later capture, recall, graph, and Cortex actions.

NAVIGATION

- Drag orbit; Shift-drag pan.
- Scroll or +/- changes depth.
- Enter/Space opens or focuses.
- Escape/Back moves outward.
- F fits; R resets.

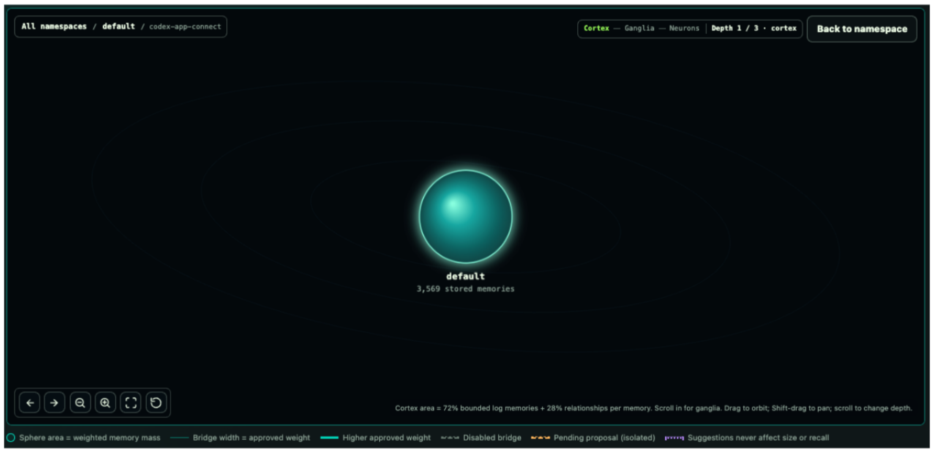
03 - CORTEX AND GANGLIA

Inside one namespace

READ-ONLY ATLAS

Scroll inward to change semantic level of detail. The namespace remains the same; only the projection changes.

“Cortex view of my “default” namespace.”



DEPTH 1 / 3 - CORTEX

“Ganglia view”

DEPTH 2 / 3 - GANGLIA

CORTEX VIEW

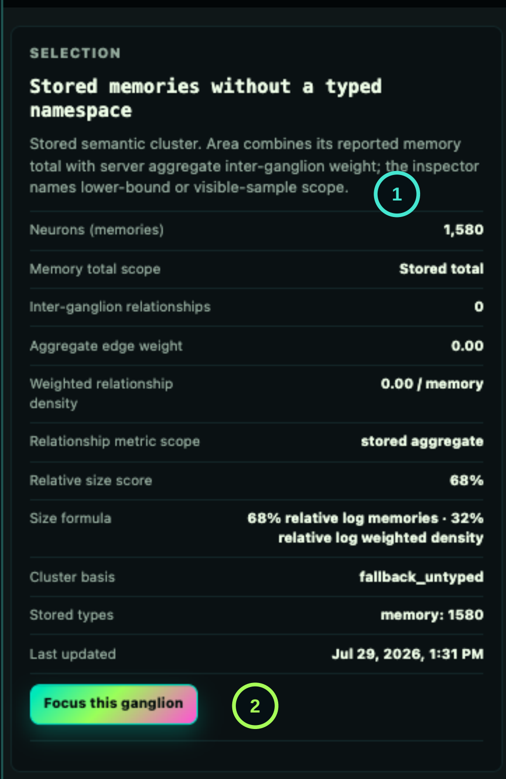
- One sphere summarizes the selected namespace.
- Area = 72% bounded log memories + 28% relationships per memory.
- Use this layer to confirm the context boundary before working.
- Scrolling inward reveals ganglia; nothing is rewritten.

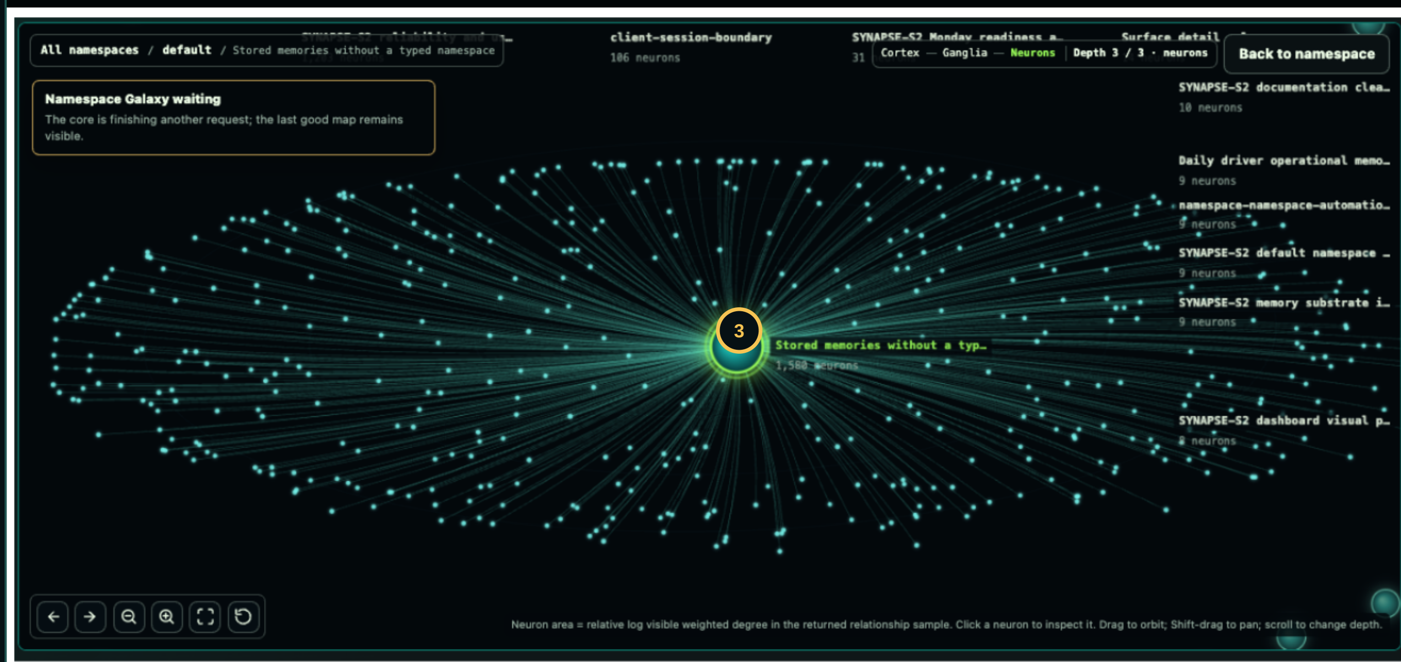
GANGLIA VIEW

- Each sphere is a derived semantic/type cluster.
- Area = 68% relative log memories + 32% weighted edge density.
- Clusters recompute from stored metadata and relationships.
- There is no direct create, rename, merge, reassign, or delete ganglion action.

What “Focus this ganglion” does

READ-ONLY





FOCUS CHANGES

Selection, camera, zoom, bounded detail request, breadcrumbs, and browser history. It centers the ganglion, resets pan, raises zoom to neuron depth, and fetches a bounded node/edge sample.

FOCUS DOES NOT CHANGE

Memory, relationships, confidence, recall scope, bridges, acknowledgement, or retention. Outside-cluster nodes may remain dimmed only for visual orientation.

1 Derived cluster detail

The inspector reports stored membership, density, basis, types, and last update.

2 The Focus action

Loads a bounded neuron projection. It is a navigation command, not a memory command.

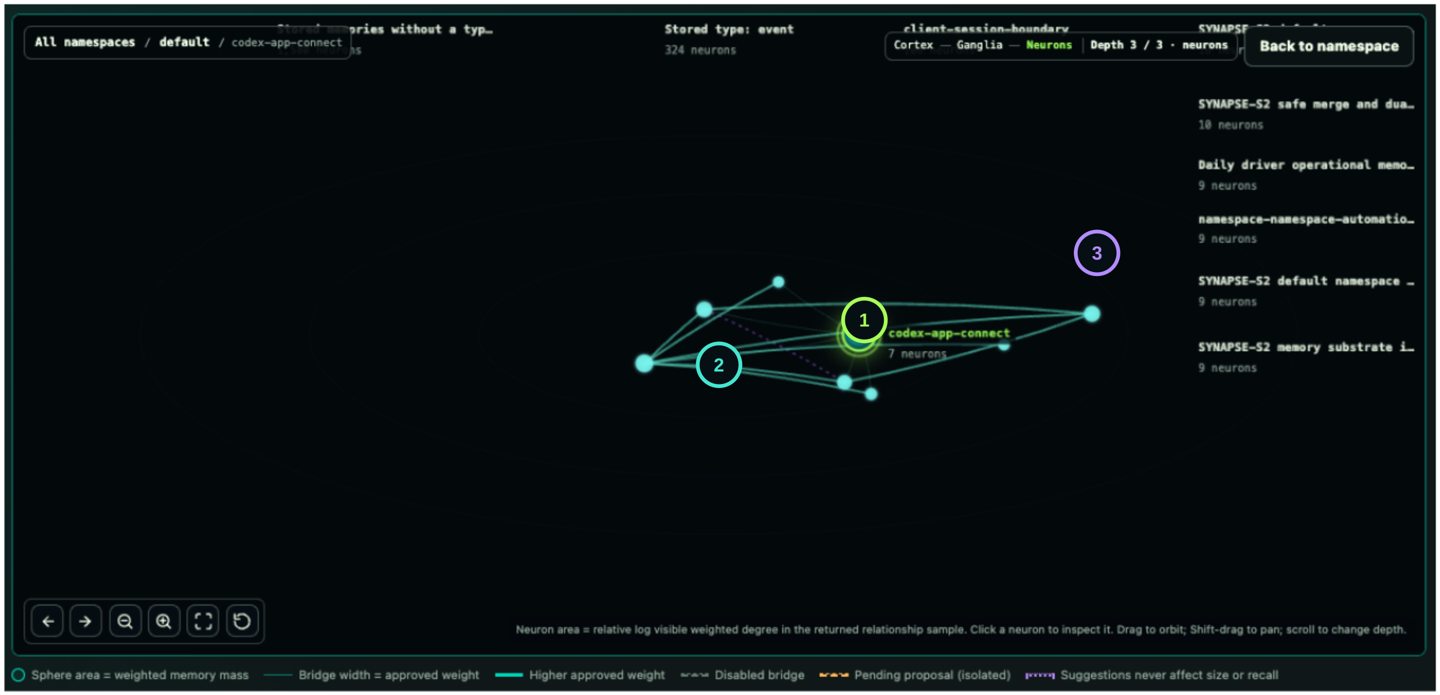
3 Focused center

The chosen ganglion becomes the center of the neuron projection.

Neuron view: inspect durable memory

READ-ONLY INSPECTOR

“Neuron view”



1 Selected neuron

One durable memory entry. The inspector can show type, tag, source, time, excerpt, and returned relationships.

2 Visible edge sample

Lines show relationships returned for this bounded projection. Visible degree is not guaranteed to be global degree.

3 Relative node size

Neuron area reflects relative log visible weighted degree inside the returned sample.

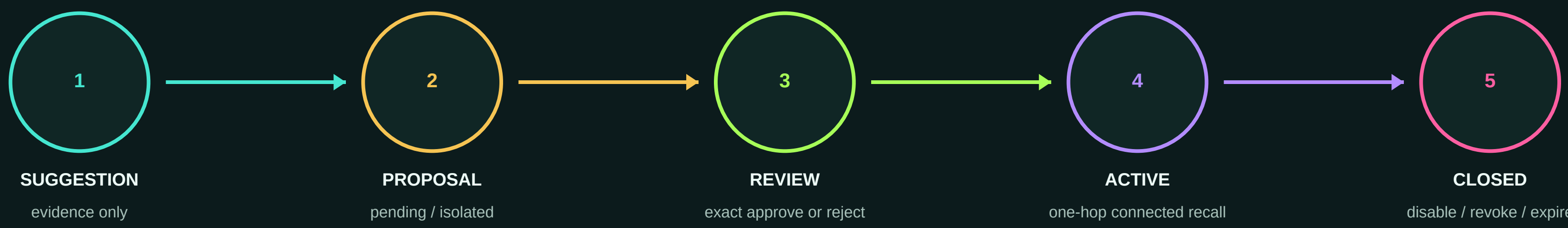
TO CHANGE A NEURON

- Promote/demote only a governed Cortex trace.
- Capture a corrected replacement when facts changed.
- Prune the exact bad node or edge in Memory Graph.
- Create recovery evidence before broad deletion.

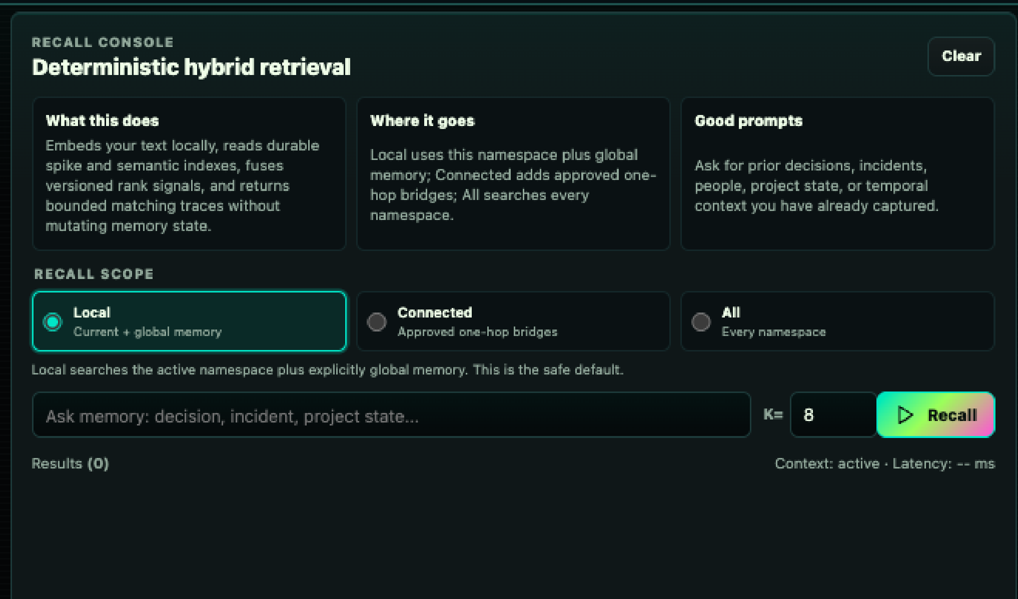
Connect namespaces with governed bridges

GOVERNED WRITE

BRIDGE LIFECYCLE



Only stage 4 grants recall authority. Suggestions and proposals never expand recall. Rejected, revoked, and expired states preserve audit history.



RECALL SCOPES

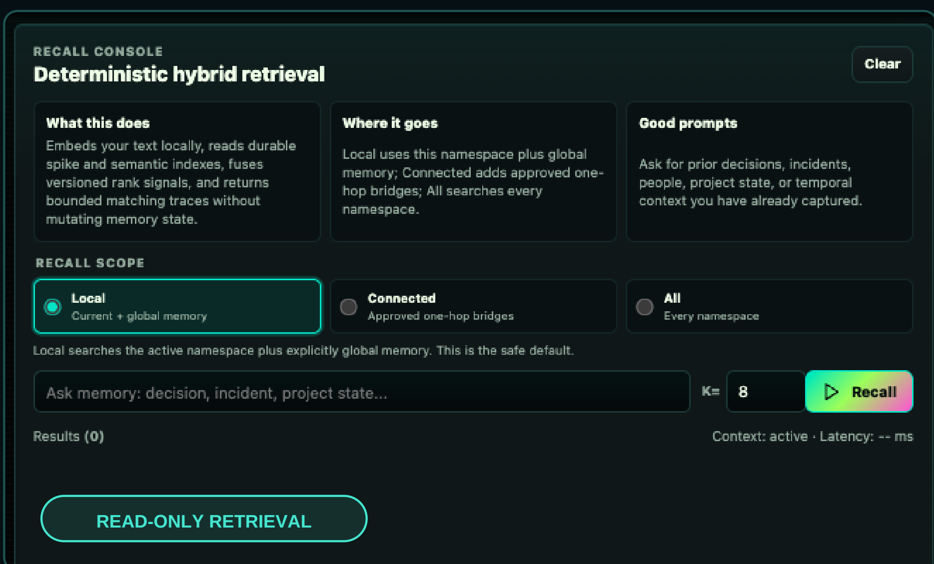
- **Local:** active context plus explicit global memory. Safe default.
- **Connected:** Local plus approved enabled one-hop bridges.
- **All:** every namespace for one explicit read-only search.

HARD GUARANTEES

- No cross-namespace memory copy or synchronization.
- No wildcard, multi-hop, or automatic approval.
- Bridge width/opacity visualizes approved weight only.
- Changing weight or direction requires a new reviewed proposal.

Capture and retrieve deliberately

DURABLE WRITE + READ



CREATION PATHS

- **Remember + publish:** one durable trace plus context update.
- **Ingest + publish:** structured events and relationships.
- **Capture conversation:** exactly-once bounded capture.
- **App Connect:** preview local UI text, then explicitly snapshot or capture a selection.

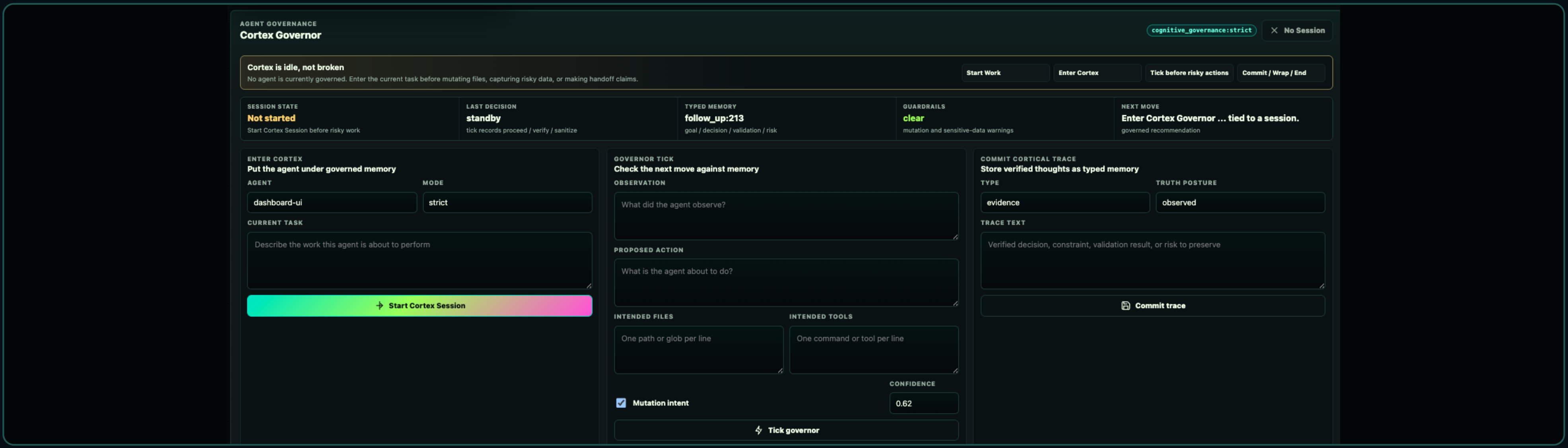
RETRIEVAL CONTRACT

Retrieval v2 is deterministic and read-only. It does not run recurrent updates, STDP, pruning, activity mutation, or cache mutation.

Always verify the active sidebar context before writing. That context determines where durable memory is stored.

Cortex Governor: govern the work

GOVERNED WRITE



FOUR-STEP LIFECYCLE

1. **Start**

   agent, mode, current task

2. **Tick**

   observation, action, files/tools, mutation intent

3. **Commit**

   typed trace plus truth posture

4. **Close**

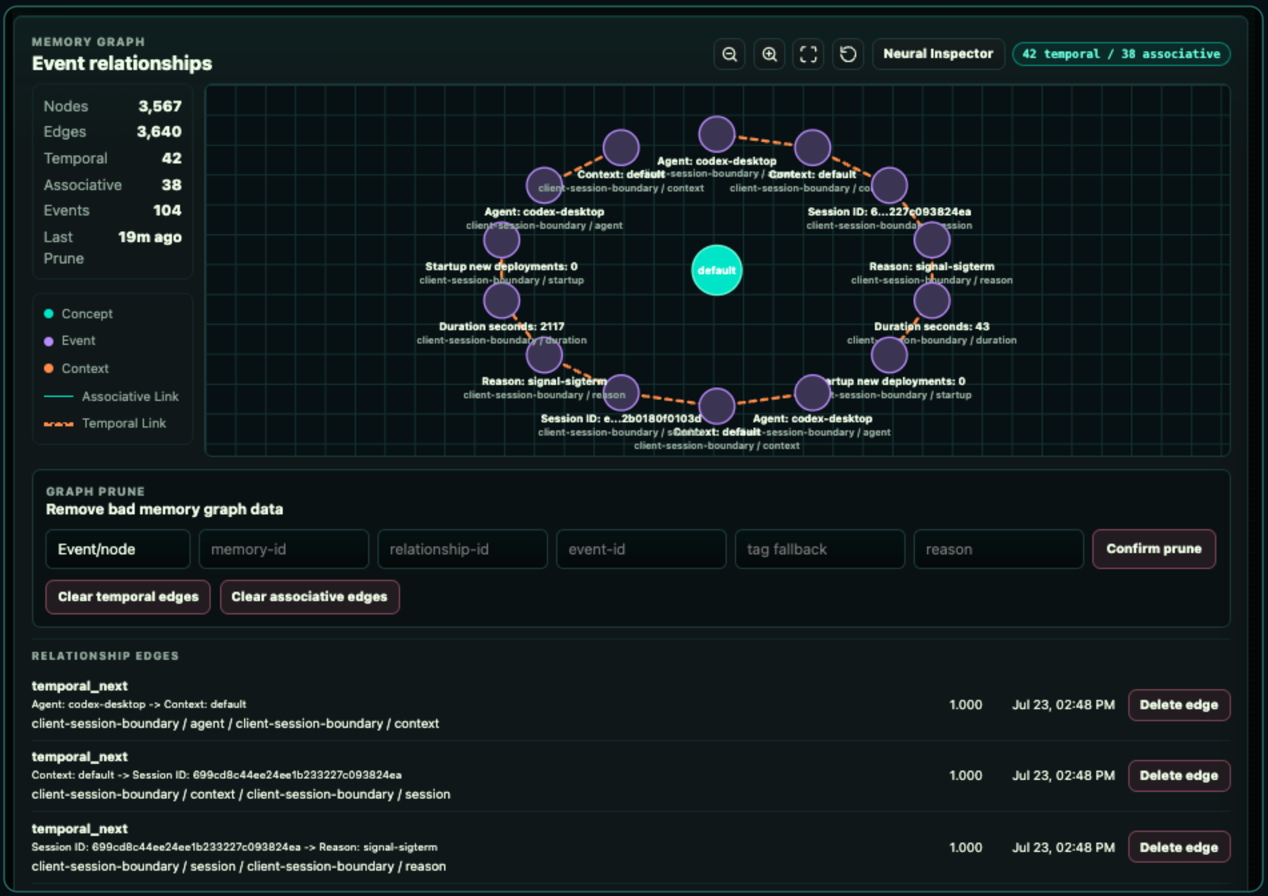
   end session or wrap a handoff

MODERATION

Promote raises governed confidence; **Demote** marks stale; **Prune** deletes the governed trace after confirmation.

Manage memory at the exact blast radius

DESTRUCTIVE CONTROLS



SUPPORTED PRUNE TARGETS

- One memory or event node.
- One exact relationship edge.
- One context / deployment event.
- All temporal edges in the context.
- All associative edges in the context.

Dragging a graph node changes only its screen position. It does not persist or rewire relationships.

SMALLEST CHANGE FIRST

Ganglia cannot be deleted directly because they are derived. Correct or prune exact member neurons and edges; the cluster recomputes.

SAFE DESTRUCTIVE CHECKLIST

1 Confirm context

2 Inspect ID + provenance

3 Create paired recovery

4 Prefer one node/edge

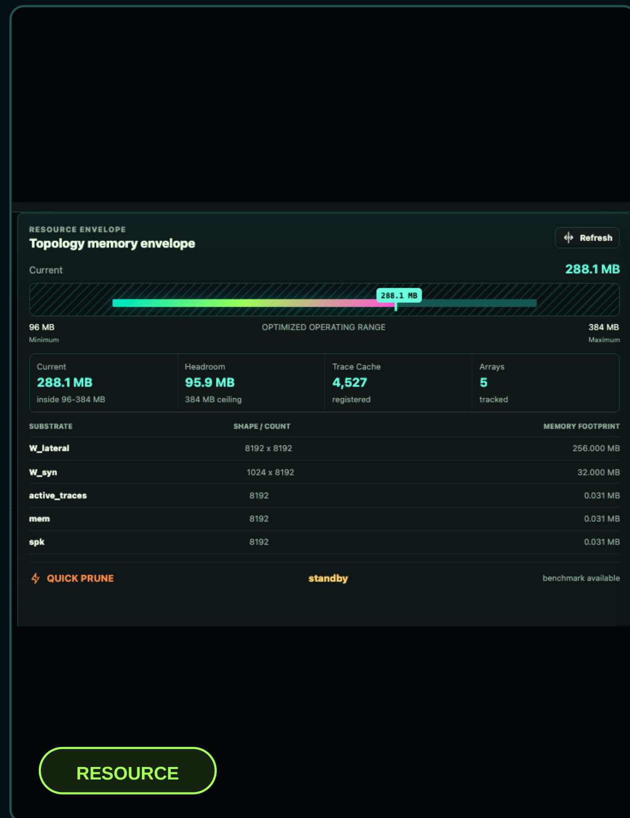
5 Give a reason

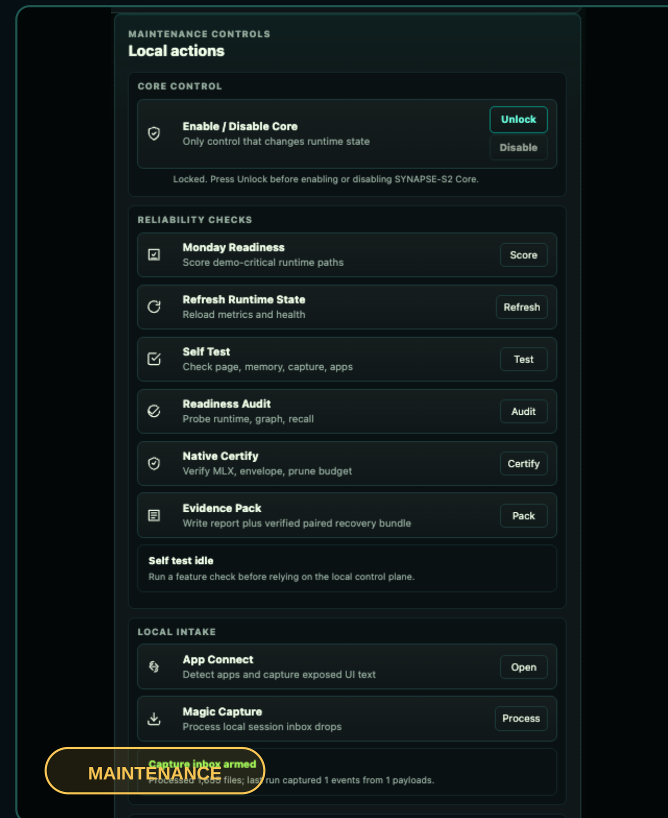
6 Confirm once

7 Doctor + recall

Reliability is a set of independent gates

DIAGNOSTIC + FILESYSTEM





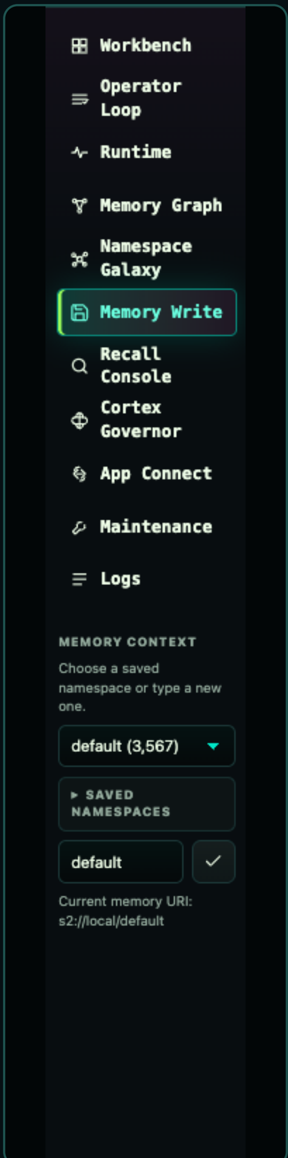
DO NOT COLLAPSE THESE INTO ONE GREEN LIGHT

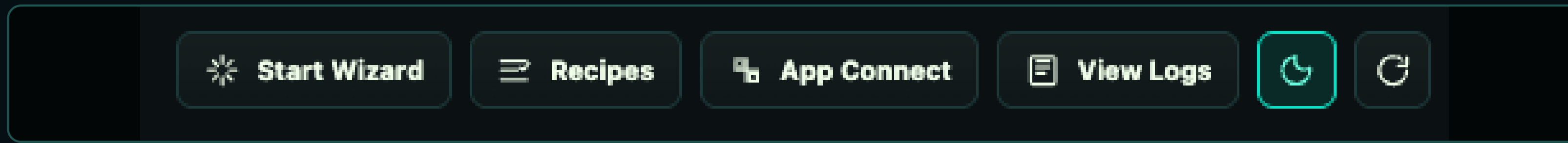
Runtime	Integrity	Delivery	Capture	Recovery
Core status and API availability.	SQLite, foreign keys, graph, and embeddings.	Lease, render, and exact receipt acknowledgement.	Inbox, exactly-once ledger, and unresolved debt.	Paired database/capture evidence and restore proof.

Quick Prune and Deep Sleep change transient neural state; neither directly deletes durable SQLite memories.

Find your way around the operator surface

OPERATOR ORIENTATION





DAILY OPERATOR LOOP

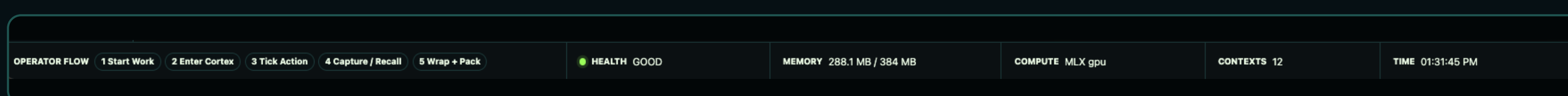
1 Choose context → 2 Start Work → 3 Enter Cortex → 4 Tick risky action → 5 Capture / Recall → 6 Inspect / correct → 7 Wrap + protect

GUIDED HELP

Start Wizard teaches the core workflow in real time. **Recipes** provides repeatable operations. **App Connect** captures exposed local UI text. **Logs** provides runtime evidence.

KEYBOARD AND POINTER CONTROLS

Drag orbit	**Shift-drag** pan	**Scroll / +/-** zoom	**Arrows** select
Enter / Space enter or focus	**Esc / Back** move outward	**F** fit	**R** reset



What operators can do - and where

CAPABILITY MAP

Intent	Surface	Effect	Class
Browse / enter	Galaxy	Select and change active scope	Read / scope
Focus ganglion	Galaxy detail	Load bounded neuron projection	Read-only
Inspect neuron	Neuron inspector	Show provenance, excerpt, visible edges	Read-only
Recall	Local / Connected / All	Deterministic retrieval	Read-only
Store / ingest	Memory Write	Create durable traces and relationships	Write
Govern work	Cortex Governor	Start, tick, commit, close	Governed
Adjust trace	Cortex card	Promote / demote governed trace	Governed
Delete exact item	Memory Graph	Prune node, edge, or event	Destructive
Connect contexts	Bridge review	Propose and approve one-hop recall	Governed
Protect state	Recovery / Evidence	Paired verified artifacts	Filesystem

CURRENT BOUNDARIES

- No direct ganglion CRUD or neuron reassignment.
- No in-place stored-text editor or topology/weight canvas.
- No namespace rename, archive, or delete lifecycle.
- Bridges are namespace-level, not neuron or ganglion-level.
- No in-place approved bridge editing.
- Dashboard cannot disable/revoke a bridge or show full history.
- Promote/Demote applies only to governed Cortex traces.
- Quick Prune does not delete durable neurons.

These boundaries keep the atlas explainable and non-mutating. Durable changes pass through explicit, auditable surfaces.

FIVE-MINUTE PAPER DEMO

1. **Galaxy** — Choose a context
2. **Cortex** — Confirm the boundary
3. **Ganglion** — Focus a cluster
4. **Neuron** — Inspect one memory
5. **Recall** — Compare recall scopes
6. **Governor** — Show controlled mutation
7. **Recovery** — Close with evidence